PANKAJ JHAJHRIA

AI / Machine Learning Engineer | Generative AI & Agentic Systems

Kurnool, AP, India | +91 7023051861 | pankajlilo252@gmail.com | github.com/Pankaj-Jhajhria | linkedin.com/in/pankaj-jhajhria

PROFESSIONAL SUMMARY

AI & Data Science undergraduate at IIITDM Kurnool with hands-on experience designing, evaluating, and deploying practical GenAI, multimodal computer vision, and machine learning systems. Proven ability to build verified RAG architectures with strict citation guardrails, multimodal segmentation benchmark apps, and large-scale data pipelines evaluated on millions of records with clean engineering rigor.

TECHNICAL SKILLS

Generative AI & Agentic Systems: RAG Pipelines, FAISS Vector Search, Dense Embeddings (Sentence-Transformers), Prompt Engineering, Source Grounding, Hallucination Guardrails, Groq API, Whisper STT, Piper ONNX TTS

Machine Learning & Data: scikit-learn, DuckDB, NumPy, pandas, SciPy, Feature Engineering, Class Imbalance Handling, Model Evaluation (Precision, Recall, F1, MRR, Recall@k)

Deep Learning & Computer Vision: PyTorch, Torchvision, CNNs, Dual-Stream ResNet, Multimodal Sensor Fusion, Semantic Segmentation, MediaPipe, OpenCV, LSTM

Software & Development Tools: Python, C++, SQL, C, FastAPI, Streamlit, Git/GitHub, Linux, REST APIs, Jupyter, pytest, SDLC

PROJECTS

MuseGuide — AI Museum Guide Using RAG & Vector Search | *Python, Groq Cloud LLM, Sentence-Transformers, FAISS, Streamlit* [Code]

- Engineered an end-to-end RAG docent indexing **2,000 CC0 museum artworks** from Cleveland Museum of Art API using normalized 384-dim Sentence-Transformers and FAISS inner-product vector search with dual generation modes (Groq Cloud LLM + deterministic offline synthesizer).
- Built confidence-threshold gating (0.35) achieving **100% negative refusal accuracy** on unsupported queries; verified benchmark evaluation with **100.0% Recall@5, 0.944 MRR, and 14.7 ms steady-state retrieval latency** with 1:1 citation parity over 38 automated pytest tests.

RGB-Thermal Semantic Segmentation & 4-Architecture Benchmark App | *PyTorch, ResNet-18, Multimodal Sensor Fusion, Streamlit* [Code | Live App]

- Implemented an RTFNet-inspired dual-stream ResNet-18 in PyTorch fusing aligned RGB and thermal features across 5 stages on the MFNet dataset (1,569 pairs, 9 classes), resolving severe 92% background class imbalance.
- Evaluated test-split performance at **46% 9-class mIoU and 40% foreground mIoU** (81% Car IoU, 58% Person IoU); built and deployed a live Streamlit comparison app benchmarking **4 fusion architectures** side-by-side with overlay inspection and ground-truth validation.

ReviewRadar — High-Throughput Sentiment Analysis & Data Pipeline | *Python, DuckDB, scikit-learn, TF-IDF* [Code]

- Architected an out-of-core data and NLP pipeline processing **5,000,000+ Amazon reviews (2.36 GB)** using DuckDB SQL joins, text normalization, and 100,000-feature TF-IDF representations to model consumer sentiment.
- Trained and calibrated cost-sensitive logistic models with tuned class weights; evaluated on an independent test split of **2,043,926 reviews**, achieving **86.99% accuracy, 0.879 weighted F1-score**, and 0.813 negative-class F1.

FilmSense — Content-Based Movie Recommendation Engine | *Python, pandas, scikit-learn, SciPy, Streamlit* [Code | Live App]

- Constructed a content-based recommendation engine vectorizing 3,911 films into 27,548 TF-IDF vocabulary terms via a priority-based metadata coalescing pipeline across heterogeneous TMDB datasets.
- Engineered weighted composite metadata representations (genres, keywords, director, cast) and cosine NearestNeighbors lookup; deployed an interactive Streamlit app supporting exact, partial, and fuzzy title matching (difflib).

EXPERIENCE

CoolieMate — Founding Team Member (cooliemate.in) *Jan 2025 – Present*

- Co-founded a digital platform for pre-booking licensed railway station porters, piloted in the Kurnool Railway Station (KRNT) market to streamline transparent passenger logistics.
- Led grassroots customer discovery, operational testing, and partner outreach, securing initial organic paying bookings (~1 booking/week pre-campaign) to validate product demand.

EDUCATION

Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Kurnool

B.Tech in Artificial Intelligence & Data Science | **CGPA: 7.42 / 10.0** *Expected Graduation: June 2028*

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Generative AI, DBMS, Operating Systems

CERTIFICATIONS & KEY HIGHLIGHTS

Deloitte Data Analytics Job Simulation (Forage) — Nov 2025: Completed practical simulation applying data analytics, forensic data validation, and exploratory visualization using Python and SQL.